

ApexPlanet Data Analytics Internship

Task 1: Data Immersion & Wrangling

Task: Task 1 – Data Immersion & Wrangling

Submitted By: P. Jeevana

Internship: ApexPlanet Software Pvt. Ltd.

Submission Date: 15/07/2026

1. Objective

The objective of this task is to understand the provided dataset, evaluate its quality, identify missing or inconsistent values, perform necessary data cleaning, and prepare a clean dataset for further analysis. This task also involves creating a data dictionary and documenting the entire data preparation process.

2. Dataset Overview

The dataset used for this task is **ApexPlanet_DataAnalytics_Dataset.xlsx**.

Dataset Details

- Number of Rows: 1000
- Number of Columns: 12

Dataset Columns

- Order_ID
- Order_Date
- Customer_ID
- Customer_Name
- Age
- Gender
- City
- Product
- Category
- Quantity
- Unit_Price
- Total_Sales

3. Tools Used

The following tools and technologies were used to complete this task:

- Python
- Jupyter Notebook (Anaconda)
- Pandas
- NumPy
- Microsoft Excel

4. Data Exploration

The dataset was explored using various Pandas functions.

The following operations were performed:

- Displayed the first five rows using `df.head()`
- Checked dataset dimensions using `df.shape`
- Displayed column names
- Examined data types using `df.info()`
- Generated summary statistics using `df.describe ()`
- Checked missing values
- Checked duplicate records

5. Data Quality Assessment

During the exploration phase, the following observations were made:

- Dataset contains 1000 rows and 12 columns.
- The Age column contained 20 missing values.
- The City column contained 13 missing values.
- No duplicate records were found.
- The Order_Date column was stored as text (object) and required conversion to datetime format.

6. Data Cleaning Process

The following cleaning steps were performed:

Handling Missing Values

- Missing values in the Age column were replaced using the median value.
- Missing values in the City column were replaced using the mode value.

Date Conversion

- The Order_Date column was converted from object type to datetime format using Pandas.

Duplicate Check

- Duplicate records were checked.
- No duplicate records were found.

Data Validation

After Cleaning:

- No missing values remained.
- All records were valid.
- Data types were verified.

7. Feature Engineering

Additional features were created to make future analysis easier.

The following columns were added:

- Order_Month
- Order_Year
- Order_Quarter

These columns were extracted from the Order_Date column.

8. Data Dictionary

A separate Data_Dictionary.xlsx file was created.

The data dictionary contains:

- Column Name
- Data Type
- Description
- Business Relevance

This document helps users understand the purpose of every variable in the dataset.

9. Final Output

The following files were generated during the task:

- Task1_Data_Wrangling.ipynb
- ApexPlanet_DataAnalytics_Cleaned.xlsx
- Data_Dictionary.xlsx
- Task1_Report.docx
- Task1_Report.pdf

10. Conclusion

The dataset was successfully explored, cleaned, and prepared for analysis. Missing values were handled appropriately, the Order_Date column was converted into the correct format, duplicate records were verified, and additional time - based features were created. The cleaned dataset and data dictionary are now ready for use in the next phase of the project.